TABLE 8. External validity—bank-level trust bias and corporate exposures.

Panel A Outcome →	(1) Corporate exposure (dummy)	(2) Corporate exposure (dummy)	(3) Corporate exposure (dummy)	(4) Corporate exposure (dummy)	(5) Corporate exposure (dummy)
Bank-level trust bias	2.200***	1.592***	1.687***	1.596***	1.556***
	[0.131]	[0.435]	[0.462]	[0.436]	[0.484]
Panel B Outcome →	Corporate exposure (log nominal)	Corporate exposure (log nominal)	Corporate exposure (log nominal)	Corporate exposure (log nominal)	Corporate exposure (log nominal)
Bank-level trust bias	18.338***	12.688***	13.868***	13.034***	12.751***
	[0.916]	[3.197]	[3.504]	[3.194]	[3.498]
Control for bank-level branches	No	No	Yes	Yes	Yes
Control for bank-level branches (squared)	No	No	No	Yes	Yes
Bank × time FEs	Yes	Yes	Yes	Yes	Yes
Target-country × time FEs	Yes	Yes	Yes	Yes	No
Home-country × target-country FEs	No	Yes	Yes	Yes	No
Home-country × target-country × time FEs	No	No	No	No	Yes
Observations	18,255	18,255	18,255	18,255	16,620

Notes: The table summarizes the results of equation (1) estimated over the full sample period from 2010-Q1 to 2021-Q2. Dependent variable is *Corporate Exposure*, defined either as a dummy variable indicating any positive exposure (panel A) or the logarithmic—log $(x + 1)$—nominal exposure (in millions; panel B) of a bank toward a target country at a point in time reported in EBA and CEBS disclosures. *Bank-level Trust Bias* is computed for each bank–target-country pair as the branch-weighted average (see equation (2)) of the residuals from a gravity model of trust (see equation (3)), where trust is defined as the portion of individuals in home country expressing "a lot of trust" toward target country, measured via Eurobarometer surveys. *Bank Branches* measures the number of bank branches (in thousands) that the bank owns in the target country. All branch-related information is from SNL Financial. For the detailed construction of the data, see Section 3. Robust standard errors are clustered at the bank level and reported in brackets. *Significant at 10%; **significant at 5%; ***significant at 1%.

widely diversified across countries are less likely to allow trust biases to affect their lending decisions. An interpretation is that more widely diversified banks are more sophisticated and have more sources of hard information. These findings are thus consistent with previous evidence that sophisticated investors are less likely to exhibit cultural biases (Grinblatt and Keloharju, 2001). In addition, trust appears to be less important—albeit insignificantly—for target countries whose bonds are frequently present in bank portfolios (panel C), an example being Germany. This is consistent with evidence that familiarity may mitigate the role of trust in financial decisions (Pursiainen, 2022).

5.6. Salience Shocks: Eurozone Crises and Brexit Referendum

In this section, we focus on two types of salience shocks potentially amplifying the role of trust in banks' lending decisions to governments. The first one is the

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Eurozone debt crisis impacting countries such as Greece and Italy. Second is the Brexit referendum, which highlighted questions about the steadfastness of the UK's commitment to the European Union and potentially amplified the role of trust in EU banks' lending decisions to the country. We think of both shocks as focusing the attention of bank investment committees on the trustworthiness of the affected countries, thereby amplifying the effects of preexisting trust biases. We thus think of these shocks as increasing the "salience" of trust biases, similar to how Fouka and Voth (2022) characterize the Eurozone crisis as increasing the salience of preexisting resentment in Greece over historic German wartime acts, and how Guiso and Makarin (2020) show that trust had a larger than average impact on trade flows following the Global Financial Crisis.

We analyze these events both by focusing on short time intervals around the relevant shocks (plus/minus 2 years) and by utilizing our full sample, estimating the following model:

$$\begin{aligned}\text{Sovereign Exposure}_{b,h,c,t} = {} & \beta_1 \text{Bank-level Trust Bias}_{bc} \times \text{Eurozone Crises}_{ct} \\ & + \beta_2 \text{Bank-level Trust Bias}_{bc} + \beta_3 \text{Bank Branches}_{bc} \\ & + \beta_4 \gamma_{bt} + \beta_5 \lambda_{hct} + \varepsilon_{bhct}, \end{aligned} \tag{4}$$

where $\textit{Eurozone Crises}_{ct}$ is a dummy variable indicating when target country c experiences a sovereign debt crisis at time point t. To gauge the point of crisis, we use a threshold for the preceding 3 month-period of at least 400 basis points average daily bond yields above that of Germany (as in Brutti and Saure, 2016).

Results are reported in Online Appendix Table D.6. Panel A defines the dependent variable as a dummy for sovereign exposures, while panel B uses the continuous version of the same variable. Columns (1) and (2) employ the full sample, whereas columns (3) and (4) create an event study setting focusing on the plus/minus two-year period around ECB President Mario Draghi's "Whatever it takes" speech on July 26, 2012. The resulting coefficients support our prediction that salience shocks strengthen the role of trust in banks' investment decisions. When faced with adverse shocks hitting sovereign debt markets of target countries, banks increasingly resort to their cultural stereotypes in deciding to which country and how much to lend. In panel A, the total effect for bank-level trust bias (interaction + baseline) is at least three times that in our baseline estimates in Table 2, underlining the importance of these time-varying salience shocks for the relationship between trust and investments.

The second part of our investigation, focusing on Brexit salience, estimates the following model:

$$\begin{aligned}\text{Sovereign Exposure}_{b,h,c,t} = {} & \beta_1 \text{Bank-level Trust Bias}_{bc} \times \text{Brexit Salience}_{hct} \\ & + \beta_2 \text{Bank-level Trust Bias}_{bc} + \beta_3 \text{Bank Branches}_{bc} \\ & + \beta_4 \gamma_{bt} + \beta_5 \lambda_{hct} + \varepsilon_{bhct}, \end{aligned} \tag{5}$$

where $\textit{Brexit Salience}_{hct}$ is a dummy variable indicating banks headquartered in EU countries (h), facing the United Kingdom as a target country (c), after Brexit

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referendum took place on June 23, 2016 (t). Note that this variable, similar to $Eurozone\,Crises_{ct}$ in our previous setting, can only be estimated in interaction form due to our saturated fixed effect specification.

Results are reported in Online Appendix Table D.7, where we again focus on both types of dependent variables (panels A and B), the full period versus event study periods around the Brexit referendum (columns (1)–(2) vs. (3)–(4)), and all exposures versus foreign exposures (columns (1)–(3) vs. (2)–(4)). Estimates for the interaction coefficient generally have the expected positive sign but are precisely estimated only in panel B. This suggests that the *Brexit Salience* operates more strongly via the intensive margin of sovereign exposures. That is, assuming Brexit was a negative shock for the UK sovereign debt market, low-trust EU banks do not exit the UK sovereign debt market more than the high-trust EU banks. Instead, they reduce their exposures.

6. Mechanisms

In this section we discuss why the cultural biases of branch employees are important for an investment strategy whose broad parameters are set by board members and officers at bank headquarters.

Decisions at bank headquarters are shaped by information and personnel flows up the organizational hierarchy from branches to the C-suite and boardroom. As the Corporate Finance Institute (2021) writes, "... banks have a rigid and strict hierarchy that is comparable to a military organization, where each rank means a great deal..." A number of studies have examined the impact of these organizational hierarchies on banks' economic decision-making. Liberti and Mian (2009) find that greater hierarchical and geographical distance between the information-collecting agent and loan-approving officer leads to less reliance on subjective information and more on objective information. Skrastins and Vig (2018) find that increased hierarchization of branches reduces the volume of credit extension, worsens loan performance, and leads to greater standardization of loan contracts. Motivation for these studies differs, but they have in common their treatment of banks as hierarchies.

We follow this literature by modeling banks as hierarchies linking headquarters, where broad strategic decisions are made, with branches and subsidiaries, from which human capital and information flow. Panel B in Figure 1 is a visual representation.[52] Loan officers, portfolio managers, investment analysts and other subordinates in

52. Despite being theoretically possible, we consider the first mechanism here, namely the delegation of sovereign bond investment decisions from bank headquarters to subsidiaries, to be of negligible importance, given that our focus is on the entry/exit decisions of multinational banks, which are centrally decided by high-level managerial teams at bank headquarters. The literature discusses cases where bank subsidiaries and branches set their own deposit rates, hire their own tellers, award promotions to their own employees, pick bank hours, and design the process for selling new investment products to retail customers (Nagar, 2002; Dlugosz et al. 2024), but not their entry/exit decisions from a particular sovereign debt market. Our discussions with individuals working in multinational banks also provided various anecdotes inconsistent with the operation of this mechanism.

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the countries in which the bank operates provide information to headquarters. Headquarters, which in practice means the CEO, board, and investment committee, is then responsible for making broad strategic decisions about the investment portfolio. Those inputs are colored by the trust subordinates display toward countries of potential investment. Those inputs are aggregated and assessed by the bank's top officers, who then establish guidelines for the bank's investment decisions.

Employees of a foreign subsidiary are residents of the country in question and tend also to be citizens of that country. This justifies imputing to them the cultural attitudes of residents of that country. In practice, the cultural stereotypes of the employees of foreign subsidiaries can influence decisions made at headquarters through disembodied information flows transmitted via internal reports, meetings, phone calls, and other types of communication. We cannot directly distinguish this channel, however, since we lack data on such internal information flows for banks in our sample.

We therefore utilize a dataset made available by Hassan et al. (2024), which documents firm-level sentiments toward various target countries expressed during earnings calls by managers of publicly traded firms. This dataset, spanning the 2002–2020 period, covers a broad range of countries where firms are located (i.e., home countries) and a broad range of countries that firm managers "talk about" (i.e., target countries). Because such communications are open to the public, managers are less likely to reveal subjective beliefs and biases in their statements, which stacks the cards against our hypothesis.

We aggregate these firm-level data at the country level and construct a dependent variable by computing the average managerial sentiments across firms located in the same home country, talking about the same target country at the same point in time. We then estimate the following specification:

$$\text{Managerial Sentiments}_{h,c,t} = \beta_1 \text{Country-level Trust Bias}_{hc} + \beta 2\gamma_{ht} + \beta_3 \lambda_{ct} + \varepsilon_{hct}. \tag{6}$$

A positive coefficient for β_1 would be consistent with the information channel in panel B of Figure 1. It implies that managers express sentiments that are positively related to the cultural trust biases of the country where their firms are located. The fact that we include time-varying target-country fixed effects makes it unlikely that managerial sentiments reflect any fundamental information regarding these target countries.

Table 9 reports the results for four alternative definitions of trust biases. The first two columns focus only on financial firms, whereas the latter two columns aggregate the data across all firms available in Hassan et al. (2024). Columns (1) and (3) include all country-pair observations, while columns (2) and (4) exclude the observations where home and target countries overlap. Overall, there is a strong positive relationship between country-level trust biases and managerial sentiments expressed with regard to different target countries. According to column (1) of panel A, a one-standard-deviation increase in trust biases is associated with a 0.16 more positive

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